

Douglas Finamore, Ph.D.

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About me

I'm a mathematician working in the fields of Dynamical Systems, Contact Dynamics, and Global Analysis. Specific research areas and mathematical skills include foliations, Lie group actions, contact dynamics, hiperbolic dynamics, billiards, and Wasserstein spaces.

Education

PhD	Universidade de São Paulo , Mathematics - GPA: 4.0/4.0 Supervisor: Dr. Carlos Alberto Maquera Apaza	Mar 2019 – Mar 2023
MS	Universidade Estadual de Campinas , Mathematics - GPA: 4.0/4.0 Supervisor: Dr. Gabriel Ponce	Mar 2017 – Feb 2019
BS	Universidade Federal de Minas Gerais , Mathematics - GPA: 3.02/4.0 Exchange year: Universitetet i Bergen, Bergen - NO	Mar 2012 – Jul 2016 Jun 2015 - Jun 2016

Experience

CMCC - UFABC , Visiting professor	Santo André, BR Jul 2026 - ongoing
IMECC - UNICAMP , Post-doctoral researcher	Campinas, BR Nov 2024 – Jul 2026
CMLS - École Polytechnique , Post-doctoral researcher	Palaiseau, FR Jan 2024 – Nov 2024
ICMC-USP , Teaching assistant Calculus I, II, and III	São Carlos, BR Mar 2020 – Nov 2021
IMECC - UNICAMP , Teaching assistant Calculus III and Advanced Linear Algebra	Campinas, BR Feb 2018 – Nov 2018

Publications

Journal Articles

Pushforward dynamics on Wasserstein spaces and measure rigidity Douglas Finamore , André Gomes and Christian Rodrigues arXiv	Preprint, 2026
A CAT(0)-approach to marked length spectral rigidity of Sinai billiards Douglas Finamore and Martin Leguil, arXiv	Preprint, 2025
Contact foliations and generalised Weinstein Conjectures Douglas Finamore 10.1007/s10455-024-09957-w	Ann. Glob. Anal. Geom., 2024
Quasiconformal contact foliations Douglas Finamore 10.1007/s00208-023-02687-7	Math. Ann., 2024

Miscellaneous

Contact foliations: closed leaves and generalised Weinstein conjectures

Douglas Finamore

10.11606/T.55.2023.tde-30082023-163143 

PhD thesis, 2023

Entropy of pseudogroups and foliations

Douglas Finamore

10.47749/T/UNICAMP.2019.1080998 

MS dissertation, 2019

Conference talks, posters, and organisation

Talks and posters

Probability, Geometry, and Dynamics

- **Talk:** *A CAT(0)-approach to the marked length spectrum rigidity of Sinai billiards*

IMECC - UNICAMP

Nov 2025

VIII Escola Brasileira de Sistemas Dinâmicos

- **Poster:** *Topics in billiard rigidity*

IMPA

Nov 2025

Séminaire de Systèmes Dynamiques de Jussieu

- **Talk:** *Estimating the number of closed leaves for contact foliations*

IMJ-PRG - Sorbonne Université

Apr 2024

Greifswald-Marburg Joint Research Seminar

- **Talk:** *Closed orbits for contact foliations*

Online event

May 2022

First Iterations in Dynamical Systems

- **Talk:** *k -contact structures and their induced foliations: closed orbits and generalised Weinstein conjectures*

Online event

Oct 2021

X Workshop de Teses e Dissertações em Matemática

- **Talk:** *Generalised k -contact structures and their induced foliations*

ICMC - USP

Nov 2020

V Escola Brasileira de Sistemas Dinâmicos

- **Poster:** *Dynamical Complexity of Foliations: Entropy and a Theorem of Ghys-Langevin-Walczak*

ICEx - UFMG

Oct 2019

XIII Encontro Científico dos Pós-Graduandos do IMECC

- **Talk:** *Entropy of foliations and pseudogroups (in Portuguese)*

IMECC - UNICAMP

Oct 2018

VII Simpósio Nacional / Jornadas de Iniciação Científica IMPA

- **Talk:** *Representations of finite groups and applications to Quantum Physics (in Portuguese)*

IMPA

Nov 2014

XXII Semana de Conhecimento e Cultura UFMG

- **Poster:** *Shor's algorithm for factoring integers (in Portuguese)*

ICEx - UFMG

Oct 2013

Conference organisation

VI Encontro Paulista de Alunos de Dinâmica

- Marketing and organisation

IMECC - Unicamp

Jan 2020

I Encontro Paulista da Pós-Graduação em Matemáticas

- Marketing and organisation

Online event

Feb 2022

II Encontro Paulista da Pós-Graduação em Matemáticas

- Marketing and organisation

IMECC - Unicamp

Jun 2026

Research projects

Geometry and dynamics on Wasserstein spaces

2025 - ongoing

- This is a collaborative project with Drs. Christian Rodrigues, André Gomes, and Luiz San Martin, developed within the research group *Geometry and Probability in Dynamical Systems* at the Max Planck Institute. My focus is on the geometry of the space $\mathcal{P}(X)$ of probability measures on the metric space (X, d) , equipped with the Wasserstein metric w , and its implications for dynamical systems. We investigate metric rigidity questions in the isometry group $\text{Iso}(\mathcal{P}(X), w)$, as well as applications of the Wasserstein metric to continuity problems for Lyapunov exponents, to the study of automorphisms defined by *pushforwards*, and to other potential uses of the geometric structure of $(\mathcal{P}(X), w)$ in ergodic theory.
- *Role*: Researcher.

Rigidity of billiards

2024 - ongoing

- This project is a collaboration with Dr. Martin Leguil (École Polytechnique) and extends our work from my postdoctoral stay at CMLS. Broadly, we ask how much information about a hyperbolic billiard can be recovered from periodic data. Specifically, we investigate under what conditions spectral rigidity holds for Sinai billiards: if two billiard tables share the same marked length spectrum, are they necessarily isometric? To answer this, we study the coarse geometry of the phase space of Sinai billiard flows and the extent to which classical rigidity results, such as those of Otal and Croke for negatively and nonpositively curved surfaces, remain valid in the CAT(0) setting.
- *Role*: Main researcher.

q -contact structures: geometry, dynamics, and applications

2023 - ongoing

- This project is a natural extension of the themes I worked on during my PhD, and focuses on the study of q -contact structures, objects that generalize classic contact structures, but allowing for codimension higher than one. As a consequence, such structures naturally define actions of the q -dimensional Euclidean space on their ambient manifolds, whose orbit foliation can then be seen as direct generalization of the flow of the Reeb vector field. There is a myriad of questions one can ask about such structures, most of them in the way of understanding which properties of a contact structure still hold in this generalized scenario. I'm currently concerned with problems of determining what are the interesting (and useful) invariants of such structures, of whether or not contact rigidity holds for them, and in applications of such objects to the efforts of classifying Anosov actions of higher rank groups.
- *Role*: Main researcher.

Dynamics and Topology of Intrinsically Harmonic Forms

2022 - 2023

- This project is a collaboration with Dr. Elizeu França. We study intrinsically harmonic differential forms and their impact on manifold topology. A key goal is to prove a conjectured “dual” of Tischler’s theorem: that an orientable closed n -dimensional manifold supporting a closed nowhere-vanishing $(n - 1)$ -form fibres over the circle.
- *Role*: Researcher.

Grants and awards

Max Planck Institute for Sciences Post-Doc Scholarship, Grant number 82068-24/5729

2024 - ongoing

CAPES Thesis Awards 2024 for Brazil’s best thesis on Mathematics, Probability, and Statistics, Honourable Mention

2024

CAPES Math/AmSud Post-Doc Scholarship, Grant number 88887.898617/2023-00

2024

CAPES Programa de Excelência Acadêmica (PROEX) Doctorate Scholarship, Grant number PROEX-11377206/D

2019 - 2023

CNPQ Master Studies Scholarship, Grant number 131555/2017-0

2017 - 2019

CAPES Science without Borders Scholarship, Grant number 88888.853750/2014-00

2015 - 2016

FAPEMIG Junior Researcher Grant

2013 - 2015

Skills

Languages: Portuguese (native), English (fluent), Norwegian, French (intermediate level), German, Italian (basic skills).

Coding: C#, SQL, JavaScript, Python, LaTeX, HTML.

Technologies: .NET, Visual Studio, TexWorks, Wolfram Mathematica, MATLAB, Geogebra.

Misc: Academic research, teaching, training, consultation, $\text{\LaTeX}$ typesetting, and publishing.

References

Dr. Carlos Maquera

- Av. Trabalhador São-Carlense São Carlos, SP
- cmaquera@icmc.usp 

*Department of Mathematics
ICMC - USP*

Dr. Martin Leguil

- 91128 Palaiseau Cedex, France
- martin.leguil@polytechnique.edu 

*Centre de Mathématiques Laurent Schwartz
École Polytechnique*

Dr. Christian Rodrigues

- Pça. Sérgio Buarque de Holanda, Campinas, SP
- rodrigues@ime.unicamp.br 

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